Pedro Lustosa Rege Botelho

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Education

2018–2023 **Aeronautics Institute of Technology** (ITA) – Sao Jose dos Campos, SP, Brazil.
B.Sc. in Electronic Engineering — Brazil's top engineering school (acceptance rate 0.98%).
○ Dissertation: "ASAGYM: A Python Library for Reinforcement Learning in Aerospace Operational Scenarios"
○ Advisor: Marcos Maximo

Research Experience

2026–Present **Research Assistant**, Carnegie Mellon University – Pittsburgh, PA, US.
Advisor: Prof. Guannan Qu, Department of Electrical and Computer Engineering.
○ Working on multi-agent systems and multi-agent reinforcement learning (MARL) in collaboration with Prof. Guannan Qu and his PhD student Alex DeWeese at the Department of Electrical and Computer Engineering.
○ We are specifically interested in establishing the foundational principles and mathematical guarantees for Locally Interdependent Multi-Agent Markov Decision Processes (LIMAMDP).
○ Currently working on LIMAMDP under stochastic communication modeled by log-distance path loss with log-normal shadowing.

2024–Present **Research Assistant (Visiting Student)**, Northwestern University – Evanston, IL, US.
Advisor: Prof. Igor Kadota, Department of Electrical and Computer Engineering.
○ Best Student Paper Award – WiOpt 2025 (ECE Team Wins Best Student Paper Award at 2025 WiOpt) "Fair Dynamic Spectrum Access via Fully Decentralized Multi-Agent Reinforcement Learning"
○ Designed and implemented fully decentralized MARL for dynamic spectrum access via a novel Distributional Recurrent Dueling DQN model, achieving up to 86% throughput and an average Jain's fairness index of $0.91/1.00$ across 52 wireless network settings (M=2–10 agents, N=1–10 bands), with robustness to jamming episodes and ad-hoc network topologies by leveraging localized interference patterns;
○ Derived from the first architecture, proposed a novel Neuromorphic Distributional Recurrent Dueling model that achieves an average Jain's fairness index of $0.96/1.00$ while maintaining an average throughput of 97%.

2020–Present **Research Scientist**, Center of Excellence in Artificial Intelligence (CEIA) & AKCIT – Goiania, GO, Brazil.
○ National joint research centers with more than 1,100 researchers and US$100 million in investment.
○ CEIA has contributed to NVIDIA, Meta, and Google DeepMind, such as GAIA.
○ Computer Vision: I engineered a real-time visual odometry algorithm for autonomous vehicles.
○ AI for Speech & Generative Models: Developed a diffusion model for singing-voice conversion for Moises (Music.AI), recognized as Apple's iPad App of the Year (2024);
○ Offline-to-Online RL for Robotics: Investigated methods for safe pre-deployment training for humanoid robots; establishing a comprehensive offline RL benchmark;
○ **[Present]** Continual learning: Investigating Kolmogorov–Arnold Networks (KANs) for continual learning through mathematical formalization and experiments, with promising preliminary results.

2023 **Research Intern**, Institute for Advanced Studies (IEAv), the Brazilian Air Force – Sao Jose dos Campos, SP, Brazil.
○ MARL Framework: Developed a MARL framework within ASA (FAB's C++ strategic simulator) for senior thesis, deploying agents to support High Command decision-making;
○ Algorithm Design: Engineered a DQN strategy using relative polar coordinates for state abstraction and a delayed-reward scheme, enabling retrospective learning from simulation outcomes;
○ Achieved a 91% successful engagement rate; framework being integrated into FAB's training pipeline to train 720 pilots, enhancing Brazil's air defense.

Work Experience

2024–2025 **1st Lieutenant Electronic Engineer**, Aeronautical Materials Park of Lagoa Santa (the Brazilian Air Force) – Lagoa Santa, MG, Brazil.
○ My main responsibilities included modernizing and maintaining the Brazilian Air Force's aircraft.
○ I worked on the navigation system update for the EMB-314 and the modernization of the Neiva T-25 and military Cessna 208 fleets.

2020 **Intern – Backend Developer & DevOps**, Conta Zap – São Paulo, SP, Brazil.
○ Conta Zap is a Fintech that offers banking services through a banking bot assistant on WhatsApp.
○ Backend Development: Engineered the core communication infrastructure for a WhatsApp-based banking bot, integrating legacy banking systems with IBM Watson AI and microservices using Python and Node.js.
○ Cloud Migration & DevOps: Orchestrated the complete system migration from IBM Cloud to AWS, designing a scalable architecture using Kubernetes, Docker, and automation scripts.

Publications *(* indicates equal contribution – 1st author)*

[P7] **P. L. R. Botelho***, D. S. Casanova*, A. I. S. Ferreira, and F. K. Lustosa da Silva, "We KAN't Stop Learning," manuscript in preparation.

[P6] **P. L. R. Botelho***, D. S. Casanova*, A. I. S. Ferreira, L. R. S. Gris, F. K. Lustosa da Silva, R. R. Faustino, F. S. Oliveira, and A. R. Galvao Filho, "When Does Locality Prevent Forgetting? Evaluating Continual Learning in MOS Prediction," submitted to IEEE ICASSP 2027.

[P5] A. I. S. Ferreira, **P. L. R. Botelho**, R. R. Faustino, F. K. Lustosa da Silva, D. S. Casanova, R. Maia, F. S. Oliveira, and A. R. Galvao Filho, "CAL-MOS: Bridging Layers with Adapters for Robust MOS Prediction Across Speech Foundation Models," to appear in Interspeech 2026.

[P4] R. P. Ferreira, **P. L. R. Botelho**, Y. Zhang, and I. Kadota, "Neuro-Cognitive Radios for Dynamic Spectrum Access," in Proc. of NeurIPS Workshop on AI and ML for Next-Generation Wireless Communications and Networking (AI4NextG), Dec. 2025.

[P3] J. P. A. Dantas, F. L. L. Medeiros, A. R. Samersla, **P. L. R. Botelho**, V. C. F. Gomes, S. R. Silva, Y. D. Ferreira, A. O. Arantes, M. R. C. Aquino, and M. R. O. A. Maximo, "Deep Reinforcement Learning Agents With Collective Situational Awareness for Beyond Visual Range Air Combat," IEEE Access, vol. 13, pp. 143052–143069, 2025.

[P2] Y. Zhang*, **P. L. R. Botelho***, T. Gordon*, G. Zussman, and I. Kadota, "Fair Dynamic Spectrum Access via Fully Decentralized Multi-Agent Reinforcement Learning," in Proc. of WiOpt, Linkoping, Sweden, May 2025, pp. 1–8. **[Best Student Paper Award Winner]**

[P1] G. H. Gobi* and **P. L. R. Botelho***, "ASAGYM: A Python Library for Reinforcement Learning in Aerospace Operational Scenarios," Senior thesis, Instituto Tecnologico de Aeronautica (ITA), Dec. 2023.

Awards and Honors

2026 **1st Place – Machine Learning Summer School (MLSS NYC 2026) Hackathon**, Columbia University, New York, NY, USA.
- MathDistill: an AI for Science pipeline that extracts theorems, lemmas, and hypotheses from mathematical papers, organizes them into a visual knowledge graph, and formally verifies each statement by generating Lean proofs.

2025 **Best Student Paper Award**, Linkoping, Sweden.
- 23rd International Symposium on Modeling and Optimization in Mobile, Ad Hoc, and Wireless Networks (WiOpt 2025)
- "Fair Dynamic Spectrum Access via Fully Decentralized Multi-Agent Reinforcement Learning,"

2024 **1st Place Award in the General Mechanics Course of the T-27M Aircraft (Embraer EMB-312)**
The Brazilian Air Force

2011 **Silver medal at the Brazilian Astronomy and Astronautics Olympiad (OBA)**

Extracurricular Activities

2026 **Machine Learning Summer School (MLSS NYC 2026), Columbia University** – New York, NY, USA (June 15–26, 2026)
- Selected participant (∼200 selected/∼ 1,200 applicants worldwide)

2024–2025 **LXAI NeurIPS 2024 Mentoring Program** (LatinX in AI).
- Mentoring program for early LATAM researchers with senior academics.
- Mentored by Felipe Leno da Silva (Principal Scientist at Microsoft) and Shagun Sodhani (Staff Research Scientist at Google DeepMind).

2019–2022 **ROBIO – Bionic Prosthesis** (Leader and idealizer)
- Developed an electronic hand prosthesis that interprets nerve impulses, applying AI for signal classification;
- ROBIO is currently developing solutions for the Brazilian National Healthcare System.

2018 **CasdVest** (Teacher and staff member)
- CasdVest provides tuition-free college preparatory education for students from low-income backgrounds.
- Teaching: Tutored Mathematics and Physics, conducting study sessions to bridge knowledge gaps for vulnerable students.
- Marketing & Fundraising: Served on the marketing team, organizing donation campaigns and product sales to support the NGO's financial sustainability.